Global to Local: A Scale-Aware Network for Remote Sensing Object Detection

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***Abstract*— With the wide application of remote sensing images (RSIs) in military and civil fields, remote sensing object detection (RSOD) has gradually become a hot research direction. However, we observe two main challenges for RSOD, namely, the complicated background and the small objects issues. Given the different appearances of generic objects and remote sensing objects, the detection algorithms designed for the former usually cannot perform well for the latter. We propose a novel global-to-local scale-aware detection network (GLSANet) for RSOD, aiming to solve the abovementioned two challenges. First, we design a global semantic information interaction module (GSIIM) to excavate and reinforce the high-level semantic information in the deep feature map, which alleviates the obstacles of complex background on foreground objects. Second, we optimize the feature pyramid network to improve the performance of multiscale object detection in RSIs. Finally, a local attention pyramid (LAP) is introduced to highlight the feature representation of small objects gradually while suppressing the background and noise in the shallower feature maps. Extensive experiments on three public datasets demonstrate that the proposed method achieves superior performance compared with the state-of-the-art detectors, especially on small object detection datasets. Specifically, our algorithm reaches 94.57% mean average precision (mAP) on the NWPU VHR-10 dataset, 95.93% mAP on the RSOD dataset, and 77.9% mAP on the DIOR dataset.**

***Index Terms*— Local attention pyramid (LAP), multiscale object detection, remote sensing images (RSIs), semantic information, small object detection.**

I. Introduction

With the development of deep learning, remote sensing object detection (RSOD) has gradually become a popular research direction [1], [2], [3], [4]. Many high-resolution remote sensing satellites around the world provide sufficient datasets for this task. Nowadays, RSOD has been widely applied in military and civilian applications, such as intelligent monitoring, urban planning, precision agriculture, and geographic information system (GIS) updating [5], [6], [7], [8]. Although many algorithms have been proposed, we find that RSOD remains challenging due to the complicated background, the wide-scale variation, and the existence of numerous small objects.

Different from natural scene images, remote sensing images (RSIs) are commonly captured from satellites with wide views, leading to lower proportion of foreground object(s) and drastic differences in object scales [9], where the small objects are often submerged by complex surrounding scenes [10]. The above issues are the main challenges for object detection in RSIs, which makes most algorithms for natural images not suitable to be applied directly to RSIs. We will first analyze the pipeline of object detectors for generic object detection and then discuss the main detectors for RSIs. Finally, we introduce our solution for RSOD, aiming to solve the complex background and small foreground issues.

According to the different image processing modes, the mainstream object detection algorithms are divided into two types: the one-stage solutions and the two-stage detectors. The one-stage algorithms extract features directly from images to predict categories and locations of objects. The typical methods include SSD [11], YOLO [12], [13], [14], [15], [16], RetinaNet [17], and so on, e.g., YOLO divides the image into several grid cells. Each grid cell detects the object locating inside itself and predicts the bounding boxes and category directly. The two-stage algorithms first generate the proposal regions based on the input image and then perform classification. Faster-RCNN [18] is one of the typical solutions. It designs a region proposal network (RPN) to gather proposals and applies RoI pooling for classification and regression. All of the above methods require a large number of anchor boxes, which brings extensive hyperparameters and calculations. Furthermore, the sizes of the anchor boxes need to be obtained by clustering according to the dataset, which is inconvenient. Therefore, some anchor-free algorithms have been proposed in recent years, such as CenterNet [19], CornerNet [20], FCOS [21], and YOLOX [22]. They transform object detection into key points or center points detection, without anchor parameters or proposals. For example, CenterNet predicts the center, width, and height of objects with hourglass networks. CornerNet detects a pair of corners of a bounding box and groups them to form the final detected bounding box.

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In addition, the feature pyramid network (FPN) [23] is widely used for multiscale object detection. Its improved methods, such as [24] and [25], are also intensively studied.

Given the differences between generic objects and remote sensing objects, many researchers have been working on multiscale object detection and feature fusion, targeting directly the small objects issue of RSIs. Wang et al. [26] proposed a multiple center point-based learning network (M-CenterNet) to improve the localization performance of tiny object detection. Ma et al. [10] presented a feature split–merge–enhancement network to emphasize the features of small objects in the feature maps and suppress the background information. Wang et al. [27] introduced a novel one-stage detector called the full-scale object detection network (FSoD-Net), which consists of multiscale enhancement network backbone cascaded with scale-invariant regression layers. Wu et al. [28] designed a feature-and-spatial aligned network (FSANet) to enhance the features of small objects so that they can be better located. Yi et al. [29] presented a lightweight multiscale feature fusion network (LMSFFNet) to achieve a better balance between the running speed of the network and the counting accuracy. Dong et al. [30] designed a receptive field expansion block (RFEB) to improve the detection performance of multiscale objects. In addition, some attention mechanisms have also been introduced into RSOD to improve the detection performance. Wu et al. [31] incorporated a residual coordinate attention module to reinforce the input feature map and strengthen the representability of objects of interest. Liu et al. [9] designed an enhanced effective channel attention (EECA) mechanism to boost the feature representation ability of the backbone. Tian et al. [32] proposed a relation-augmented embedded graph attention network (EGAT) to improve the detection performance. Ma et al. [33] applied a channel attention module to enhance the network's attention on ship targets and suppress background noise. Hu et al. [34] brought in deformable convolution to build a local attention module (LAM) for better obtaining local information of ships and effectively enhance the robustness of the network. Zhang et al. [35] presented the cross-scale self-attention (CSSA) module to suppress the noise and complex background and enhance the ability to detect multiscale targets.

Although the above approaches have made some progress, most of them ignore the importance of global semantic information, which plays a key role in grasping the background and objects contained in the image and the relationship between them. Moreover, some filtering methods are likely to be required for removing the interference between object features at different scales from the fused feature maps (e.g., Ma et al. [10] designed a feature split-and-merge (FSM) module to eliminate salient information of large objects to highlight the features of small objects in the shallow feature maps).

To address the abovementioned issues, we proposed a novel global-to-local scale-aware detection network (GLSANet) with a series of components to increase the detection accuracy. Initially, to diminish the disturbance of the background to the foreground objects, we design a global semantic interaction module (GSIIM). It provides a clearer distinction between the background and objects (see Fig. 1). Then, an optimized

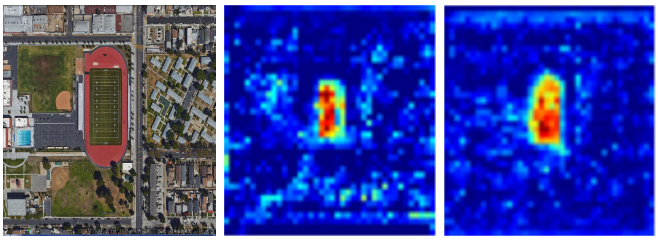


Fig. 1. Illustration of semantic information changes before and after using the proposed GSIIM. (a) Original image. (b) Deep feature map without GSIIM. (c) Deep feature map with GSIIM.

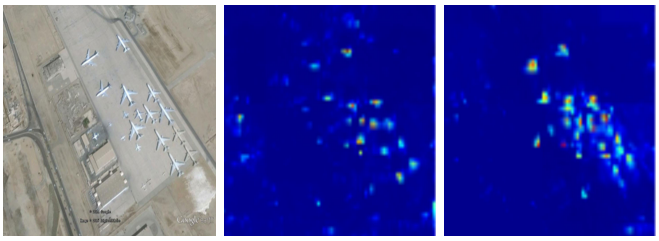


Fig. 2. Illustration of changes in small object features before and after using the proposed LAPM. (a) Original image. (b) Deep feature map without LAPM. (c) Deep feature map with LAPM.

FPN is adopted to fuse the enhanced semantic information into multiscale feature maps. Finally, we introduce the local attention pyramid (LAP) [36] to highlight the features of small objects and suppress the background and noise without extra complex filtering methods (see Fig. 2). Extensive experiments are conducted to evaluate the effectiveness of the proposed detector on three representative datasets.

We summarize our main contributions as follows.

1) We propose a novel scale-aware network to simultaneously address the issues of complex background interference and numerous small objects in RSIs via a scale-aware network with global/local feature aggregation.
2) In order to discriminate between background and remote sensing objects better, we design a plug-and-play GSIIM to excavate and reinforce the high-level semantic information in the deep feature map.
3) To improve the detection accuracy of small objects, for the first time, we introduce the LAP to enhance the features of small objects and suppress features of background and noise.

The rest of this article is organized as follows. We discuss the related techniques in Section II and present the proposed solution for RSOD in Section III. We perform extensive experiments in Section IV to verify the effectiveness of our solution. This article is concluded in Section V.

II. Related Work

A. Semantic Information Exploitation

The semantic information in object detection explains the objects and background contained in the image and the relationships between them. The earliest paper that

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shows the significance of utilizing semantic information for object detection is FPN [23]. It uses a top-down pathway to pass the semantic information in the deep feature map down and enhance the shallower feature maps at multiple scales. This greatly improves the object detection accuracy. Tian et al. [32], Fu et al. [37], and Cao et al. [38] also showed the importance of using semantic information for object detection. Therefore, it is more crucial to mine and extract semantic information for RSIs with complex background. Dai et al. [39] designed a feature fusion method by combining the bottom-to-up and up-to-bottom ways to strengthen the positioning and semantic feature of multiscale ships, particularly small ships. Cui et al. [40] applied spatial shuffle-group enhance (SSE) attention to obtain stronger semantic features of the image based on CenterNet. Zhang et al. [41] designed a compound semantic feature fusion (CSFF) to generate an effective semantic description.

However, these aforementioned methods extract and enhance the semantic information by relying on external methods, such as feature fusion or attention mechanisms. They ignore the potential of the feature map to contain information on its own. In this article, we design a novel semantic information interaction module named GSIIM to exploit and strengthen the semantic information contained in the deep feature map. Fig. 1 shows the changes of semantic information within the deep feature map before and after using the proposed module. Fig. 1(a) shows the original RSI containing a playground, Fig. 1(b) shows the deep feature map of this image without GSIIM, and Fig. 1(c) shows the feature map of it with GSIIM. It is obvious that the feature of the playground becomes robuster and clearer after we apply the proposed module. At the same time, the background interference is also diminished. Therefore, the GSIIM enhances the semantic information in deep feature maps, which alleviates the interference of complex background on foreground objects.

B. Small Object Detection in RSIs

Small object detection has always been challenging in the field of object detection for two main reasons. First, small object covers a small part of the whole image, only a few or even several pixels. Moreover, pooling operations in the backbone will further reduce the size of the feature map, which makes the features of small object may disappear or overwhelmed by the background. Second, RSIs have large and complex backgrounds, which bring disgusting but inevitable interference to the small object detection. Therefore, researchers have carried out extensive studies on small object detection. Deng et al. [42] proposed the extended FPN (EFPN) with an extra high-resolution pyramid level specialized for small object detection. Chen et al. [43] designed a parallel bifusion feature pyramid structure with a bottom-up fusion module (BFM) to detect both small and large objects at once with high accuracy. Yang et al. [44] presented a sampling fusion network (SF-Net), which fuses multilayer features with effective anchor sampling to improve the sensitivity on small objects. Wu et al. [45] inserted a squeeze and excitation block (the most classical method of channelwise attention) to focus more attention on low-level small objects. Although the above methods improve the detection accuracy of small objects from the perspective of the whole feature map, limited research has been done to enhance the features of small objects from the perspective of the small local areas where their features are located.

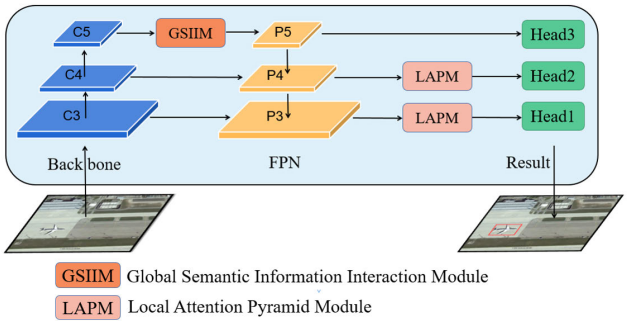


Fig. 3. Framework of our proposed GLSANet. For an input RSI, GLSANet first uses the ResNet50 backbone to extract multiscale feature maps $\{C3, C4, C5\}$. Then, the GSIIM mines and enhances the semantic information in the deep feature map $C5$ to obtain the enhanced feature map $P5$. After that, FPN fuses the abovementioned feature maps to produce shallower feature maps $P3$ and $P4$ with rich semantic information. Two LAP modules enhance the representation of small object features in $P3$ and $P4$. Finally, GLSANet uses the three strengthened feature maps for object detection.

We introduce LAP [36] to divide the feature map into multiple small parts and attempt to highlight the features of potential small objects while suppressing background and noise within each small part. Fig. 2 shows the effectiveness of this module. Fig. 2(a) shows the original image containing some small-scale airplanes, Fig. 2(b) shows its feature map without LAPM, and Fig. 2(c) shows its corresponding feature map with LAPM. We notice that the features of these small airplanes have been enhanced with the LAPM, especially those tiny ones in the center of the image. Some interference features on the left of Fig. 2(b) are also removed. Consequently, LAPM reinforces the features of small objects in RSIs and suppresses the interference of background and noise.

III. Methodology

The framework of GLSANet is summarized in Fig. 3. For the input RSI, we first employ the backbone network (ResNet50 [46]) to obtain multiscale feature maps. Then, the high-level semantic information in the deep feature map will be mined and reinforced by the global semantic information interaction module (GSIIM) (see Section III-A). After that, the enhanced deep feature map is fused with the shallower ones through a top-down path by the FPN. For the middle and shallow feature maps, two local attention pyramid modules (LAPMs) (see Section III-B) will be applied to intensify the feature representation of small objects in them step by step. Finally, the three processed feature maps are then sent to their respective detection heads (see Section III-C) for RSOD.

A. Global Semantic Information Interaction Module

Inspired by convolution [47] and self-attention mechanisms in [48], [49], and [50], we design a convenient, plug-and-play module for GSIIM. Its structure is shown in Fig. 4. Note

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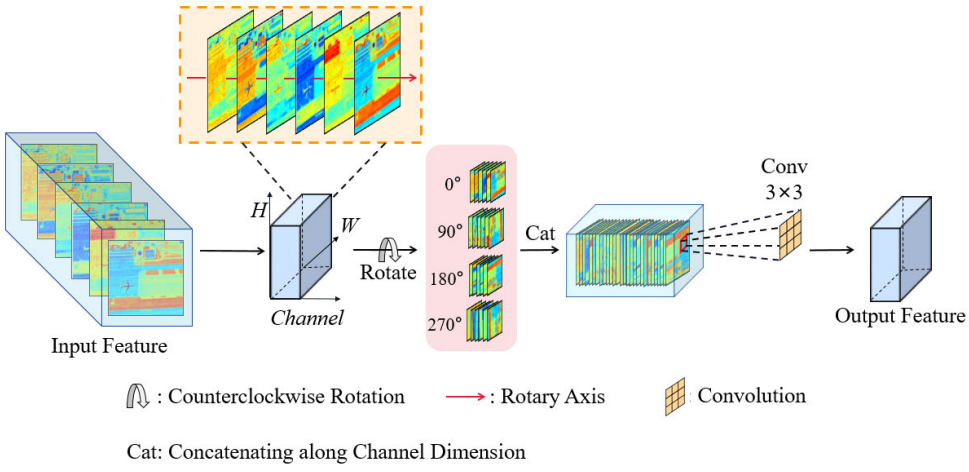


Fig. 4. Architecture of our proposed GSIIM. For an input feature map of one RSI, GSIIM first rotates it counterclockwise by 90°, 180°, and 270° around the rotation axis. Then, it concatenates the input feature map with the three rotated ones along the channel dimension. Finally, it applies a 2-D convolution to realize the interaction of global semantic information.

that, the input feature is the deep feature map obtained from one RSI through the backbone network. As shown in Fig. 3, assume that the deep feature map $C5 \in R^{N\times C\times H\times W}$, where N represents the number of batch size and C is the number of channels. H and W are the height and the width of the feature map, respectively. Thus, the data in $C5$ consist of the N feature maps of N images. To avoid the semantic information of different images interfering with each other, GSIIM will enhance the feature maps of these images independently. Denote the xth image's feature map as $C5_x \in R^{1\times C\times H\times W}$, $x \in [1, 2, \ldots, N]$. Therefore, $C5 = \{C5_1, C5_2, \ldots, C5_N\}$. Assume that the first image's feature map $C5_1$ is the input feature map of GSIIM. Then, we take the horizontal line at the position $(H/2, W/2)$ in the coordinate system where $C5_1$ is located at the rotation axis (the red arrow in Fig. 4) and rotate $C5_1$ counterclockwise by 90°, 180°, and 270°. We name the three rotated feature maps $C5_1_\alpha$, $C5_\beta$, and $C5_\gamma$ in turn. Next, we concatenate $C5_1$, $C5_1_\alpha$, $C5_1_\beta$, and $C5_1_\gamma$ along the channel dimension to get the concatenated feature map $\text{Cat}_C5_1 \in R^{1\times 4C\times H\times W}$

$$\text{Cat}_C5_1 = \text{Concat1}(C5_1, C5_1_\alpha, C5_1_\beta, C5_1_\gamma) \quad (1)$$

where Concat1$(\cdot)$ represents the channelwise concatenation.

To achieve GSIIM, we introduce a 2-D convolution. In this way, the high-level semantic information at different positions interacts through this convolution. With this interaction, we further mine out the potential semantic information in the feature map. As a result, the information of background and objects in the feature map will be stronger and clearer. Therefore, we get a new mined and enhanced feature map $C5'_1 \in R^{1\times C\times H\times W}$

$$C5'_1 = \text{Conv}_{3\times 3}(\text{Cat}_C5_1) \quad (2)$$

where $\text{Conv}_{3\times 3}(\cdot)$ means a convolution layer with 3×3 kernel size, $4C$ input channels, and C output channels. Its stride and paddings are both equal to 1. Note that, since the background occupies a large portion of the RSI, we choose a convolution with a small kernel size to interact out the features of the objects that only occupy a small portion of the image as much as possible. In fact, we can adjust the kernel size of the convolution according to the prevalent size of the objects in the images (no matter whether they are natural images or RSIs).

After the feature map of each image ($C5_x \in R^{1\times C\times H\times W}$, $x \in [1, 2, \ldots, N]$) is strengthened by the designed GSIIM, a new feature map $C5'_x \in R^{1\times C\times H\times W}$, $x \in [1, 2, \ldots, N]$, is obtained. Then, we concatenate the N new feature maps along the batch dimension. After the following processing, the complete reinforced deep feature map $P5 \in R^{N\times C\times H\times W}$ is acquired

$$P5 = \text{ReLU}\big(\text{BN}\big(\text{Concat2}\big(C5'_1, C5'_2, \ldots, C5'_x\big)\big)\big) \quad (3)$$

where Concat2$(\cdot)$ represents concatenating along batch dimension and BN$(\cdot)$ and ReLU$(\cdot)$ represent the batch normalization and the ReLU activation function, respectively.

$P5$ substitutes $C5$ in FPN to transfer semantic information downward. Considering that $C3$ and $C4$ are not abundant in semantic information, the same information interaction strategy is not performed on them. Finally, we get three output feature maps filled with richer global semantic information through the following feature fusion method:

$$P5 = \text{GSIIM}(C5) \quad (4)$$

$$P4 = \text{Upsample}(\text{Conv}_{1\times 1}(P5)) \oplus C4 \quad (5)$$

$$P3 = \text{Upsample}(\text{Conv}_{1\times 1}(P4)) \oplus C3 \quad (6)$$

where GSIIM$(\cdot)$ means the GSIIM for each image in the batch, Upsample$(\cdot)$ represents nearest neighbor interpolation with a scale factor of 2, $\text{Conv}_{1\times 1}(\cdot)$ is the pointwise convolution, which is used to adjust the number of channels, and $\oplus$ is the elementwise summation. Note that, $C5$ and $P5 \in R^{N\times 1024\times 20\times 20}$, $C4$ and $P4 \in R^{N\times 512\times 40\times 40}$, and $C3$ and $P3 \in R^{N\times 256\times 80\times 80}$ in our experiments.

We summarize the main procedure of GSIIM in Algorithm 1. In addition to mining and enhancing semantic information, it also improves the detection accuracy of

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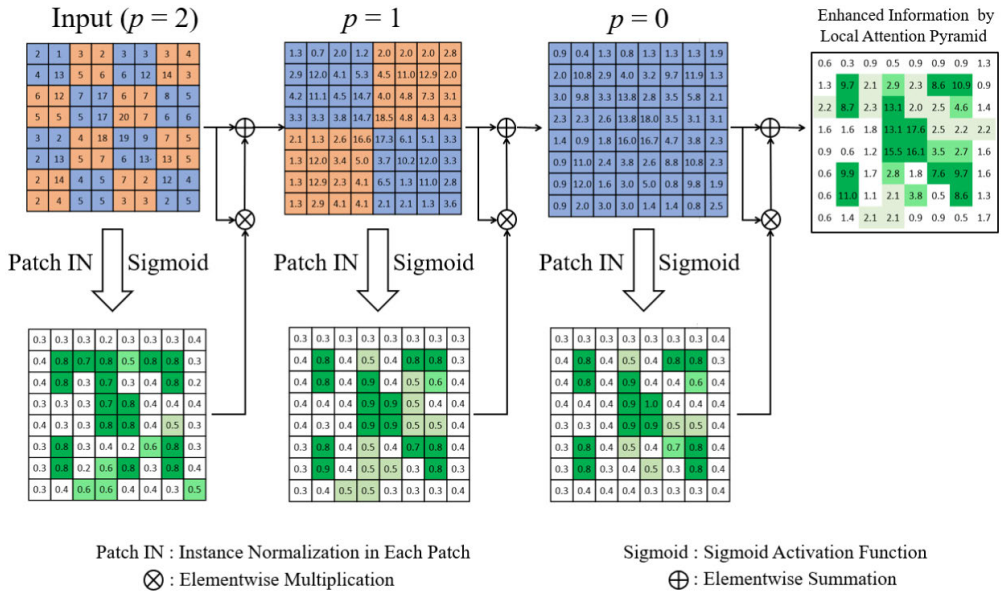


Fig. 5. Architecture of the proposed LAPM. The input feature map will be divided into different sized patches according to the level p. LAPM performs instance normalization and sigmoid within each patch independently to obtain the local attention maps. Then, it applies the attention maps to the current feature map and passes the result to the next level. LAPM encourages the model to capture and reinforce the features of small objects by amplifying locally high attention scores in every multiscale patches.

Algorithm 1 Procedure of GSIIM

Input: $X \in R^{N\times C\times H\times W}$

X refers to any one deep feature map abundant in semantic information.

Step 1: Separation.

Separate out the batches ($X_i \in R^{1\times C\times H\times W}, i \in [1, 2, \ldots, n]$) corresponding to each image in X.

Step 2: for i **in range(1, N):**

Step 2.1: Rotation.

Using the plane geometric center of the feature map as the center of rotation, rotate each batch by 90°, 180° and 270° to obtain X_i_α, X_i_β and X_i_γ, respectively.

Step 2.2: Convolution.

Concatenate X_i, X_i_α, X_i_β and X_i_γ by channel dimension and then convolve them to get $X'_i \in R^{1\times C\times H\times W}$.

Step 3: Polymerization.

Concatenate all X'_i by batch dimension to obtain the semantic information interacted feature map $X' \in R^{N\times C\times H\times W}$.

Output:

Output = ReLU(BN(X')). BN refers to batch normalization and ReLU means ReLU activation function.

large-scale objects since it works on the deep feature map (having a larger receptive field). After the above semantic information interaction and feature fusion, our network recognizes the background and objects better. The interference of background and noise to object detection is effectively minimized.

B. Local Attention Pyramid Module

According to the definition of convolution, the data of each channel in the output feature map are convolved by a corresponding convolution kernel. Since the initialization of the convolution kernel is different, the features obtained by each channel are not the same, and they all have their own focus features (having greater scores than its neighbors). Therefore, in order to enhance the features of small objects as much as possible, we introduce the LAPM [36] to enhance the feature of small objects in every channel while suppressing background and noise. The main idea of LAP is to divide each channel of the feature map into many patches and then perform normalization and sigmoid activation inside each patch independently. As a result, we can get local peaks of each patch. This enables us to spread attention to many regions of this channel at the same time. The local attention maps of all patches compose the attention map of the whole channel. Furthermore, LAP will further divide the feature maps into various sized patches based on a spatial pyramid because of the scale diversity of various objects.

The architecture of LAPM is shown in Fig. 5. The input feature map will first be at the top level after entering the pyramid (see the leftmost in Fig. 5). Then, it is propagated to the bottom of this pyramid after the feature reinforcement at each level (from left to right in Fig. 5). The output feature map of the current level is the input of the next level. The LAP will divide the feature map into patches with different sizes according to the current level p. The size of these patches is $1/2^p$ of the original, $p = n, n-1, \ldots, 2, 1, 0$ (assuming that the height and width of the input feature map are equal). The higher the level is, the larger p is, and the smaller the size of the patches is. Considering the fact that feature maps usually have multiple channels, each patch has the same number of channels as the input feature map. So how can we perform normalization on the channel dimension within each patch according to LAP? The answer is instance normalization [51], which implements normalization on every channel.

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Suppose that the input feature map is $F \in R^{N\times C\times H\times W}(H = W)$ and the highest level of the LAP is n. Then, the input feature map of the pyramid at the top level ($p = n$) is F_p. The LAPM computes the attention map $w_p \in R^{N\times C\times H\times W}$ as follows:

$$F_p = F \tag{7}$$

$$w_p = M(F_p, p) = \sigma\left(\text{PatchIn}(F_p, p)\right) \tag{8}$$

where $M(\cdot, p)$ is the calculating pth level attention map, $\sigma(\cdot)$ is the sigmoid function, and PatchIn$(\cdot,\cdot)$ means performing instance normalization within each patch. We apply the attention map to the current feature map and then pass the result to the next level as the following recursive equation:

$$F_{p-1} = \theta \cdot F_p + (1-\theta)\cdot w_p \odot F_p \tag{9}$$

where θ is a weighting factor that controls the amount of influence from previous attention scores [36] and $\odot$ denotes the elementwise multiplication. We finish the aforementioned recursion, while $p = 0$, and the patch size covers the whole size of F. F_{-1} is the final output.

In order to describe the workflow of LAPM in detail, we take the input feature map $F \in R^{N\times C\times 8\times 8}$ as an example. As shown in Fig. 5, we start the pyramid with $p = n = 2$. The patch size is 2×2 ($8/2^2 = 2$). All patches cover the whole feature map without overlap. The attention map w_2 is calculated by (7) and (8)

$$F_2 = F \tag{10}$$

$$w_2 = M(F_2, 2) = \sigma(\text{PatchIn}(F_2, 2)). \tag{11}$$

The input of level 1 is obtained by (9)

$$F_1 = \theta \cdot F_2 + (1-\theta)\cdot w_2 \odot F_2. \tag{12}$$

Following the same approach, we will acquire w_1, F_0, w_0, and F_{-1} as the following equations:

$$w_1 = M(F_1, 1) = \sigma(\text{PatchIn}(F_1, 1)) \tag{13}$$

$$F_0 = \theta \cdot F_1 + (1-\theta)\cdot w_1 \odot F_1 \tag{14}$$

$$w_0 = M(F, 0) = \sigma(\text{PatchIn}(F_0, 0)) \tag{15}$$

$$F_{-1} = \theta \cdot F_0 + (1-\theta)\cdot w_0 \odot F_0 \tag{16}$$

where w_1 and w_0 refer to the attention maps of levels 1 and 0, respectively, F_0 is the input of level 0, and F_{-1} is the output of LAPM.

Therefore, such an attention pyramid highlights the features of small objects in the feature map level by level via amplifying locally high attention scores in every multiscale patches. Meanwhile, due to the effect of attention, the background and noise are also suppressed step by step via locally low attention scores. We apply the above LAPM to the shallower feature maps $P4$ and $P3$ that are widely used to detect small-scale objects in multiscale object detection. $P4$ and $P3$ are obtained from (5) and (6). Since $P3$ has a larger size, the highest level n of its pyramid is 3 and the patch sizes are $\{10\times 10, 20\times 20, 40\times 40, 80\times 80\}$. Also, $n = 2$ for $P4$ with $\{10\times 10, 20\times 20, 40\times 40\}$ sized patches. The enhanced feature maps of them are described as the following equations:

$$P4' = \text{LAPM}(P4, n = 2) \tag{17}$$

$$P3' = \text{LAPM}(P3, n = 3) \tag{18}$$

where LAPM$(\cdot, n)$ means putting the input feature into an n-level LAP.

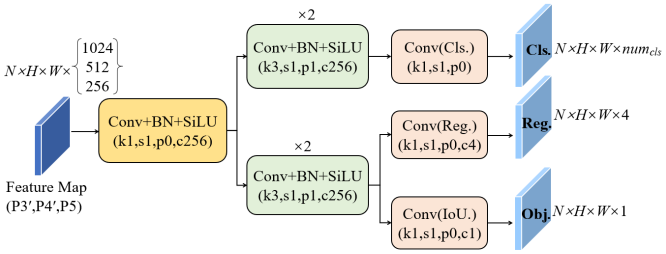


Fig. 6. Structure of the decoupled detection head. It decouples the input prediction feature map into three branches by eight convolution blocks. The three branches are used for classification, locating, and objectness tasks.

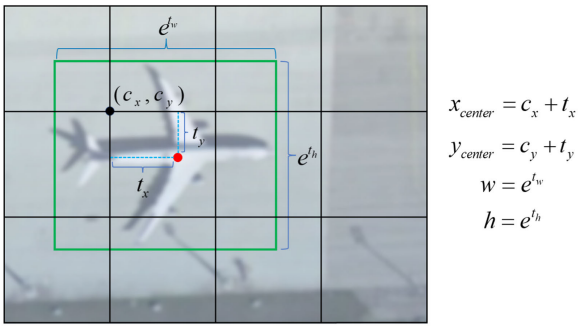


Fig. 7. Prediction box. The red point refers to the center of the object. The black point means the top-left corner (c_x, c_y) of the grid cell. t_x and t_y are offsets of the object center from the top-left corner. t_w and t_h are the width and height factors of the object, respectively. The center $(x_{\text{center}}, y_{\text{center}})$, width w, and height h of this prediction box are calculated according to the formulas on the left side of this figure.

C. Detection Heads

In object detection, the conflict between classification and regression tasks is a well-known problem [22], [52], [53]. Therefore, decoupling heads for classification and locating are widely used in most one- and two-stage detectors in recent years. Hence, our network introduces decoupling detection head as YOLOX [22]. Its structure is shown in Fig. 6. The head is composed of several convolution blocks. The first block unifies the number of channels of the input feature map to 256. Then, the data are decoupled into two branches: one for classification (Cls.) and the other for locating (Reg.) and objectness (Obj.). Objectness means the probability of object existing in one grid cell (equal to 1 if the object exists else is equal to 0). The Reg. feature map gives (t_x, t_y, t_w, t_h) for each grid cell to predict object box (shown in Fig. 7). t_x and t_y are offsets of the object center point from the top-left corner (c_x, c_y) of the grid cell. t_w and t_h are the width and height factors

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of the object, respectively. The size parameters (x_{center}, y_{center}, w, and h) of prediction box are calculated by the following equations:

$$x_{\text{center}} = c_x + t_x \tag{19}$$

$$y_{\text{center}} = c_y + t_y \tag{20}$$

$$w = e^{t_w} \tag{21}$$

$$h = e^{t_h} \tag{22}$$

where (x_{center}, y_{center}) is the center point coordinate of the prediction box in the input prediction feature map. w and h are its width and height, respectively. Note that, each input prediction feature map $\{P3', P4', P5\}$ has its own detection head whose parameters are not shared.

D. Loss Function

The total loss function of our network is defined as

$$\text{Loss} = \frac{L_{\text{cls}} + \lambda L_{\text{reg}} + L_{\text{obj}}}{N_{\text{pos}}} \tag{23}$$

where L_{cls}, L_{reg}, and L_{obj} denote the classification loss, the regression loss, and the objectness loss, respectively, and N_{pos} is the number of positive samples.

The classification loss L_{cls} is defined as

$$L_{\text{cls}} = -\sum_{i=1}^{N_{\text{pos}}}\sum_{k=1}^{C}\left(\hat{p}_{\text{ik}}\log(p_{\text{ik}}) + \left(1-\hat{p}_{\text{ik}}\right)\log(1-p_{\text{ik}})\right) \tag{24}$$

where C indicates the number of categories, $\hat{p}_{\text{ik}}$ indicates the label (0 or 1) whether the sample i belongs to class k, p_{ik} denotes the prediction category probability that the sample i belongs to class k, and $\log(\cdot)$ means the natural logarithm. The regression loss L_{reg} is calculated as follows:

$$L_{\text{reg}} = \sum_{j=1}^{S}\left(1 - \frac{\left|B_j^{\text{bd}} \bigcap B_j^{\text{gt}}\right|}{\left|B_j^{\text{bd}} \bigcup B_j^{\text{gt}}\right|}\right) \tag{25}$$

where S is the number of bounding boxes, B_j^{bd} and B_j^{gt} represent the bounding box j and its corresponding ground truth box, respectively, $\bigcap$ means intersection, and $\bigcup$ means union. The objectness loss function L_{obj} is expressed as

$$L_{\text{obj}} = -\sum_{i=1}^{N_{\text{pos}}}\sum_{l=1}^{G}\left(\hat{\text{obj}}_i^l \log\left(\text{obj}_i^l\right) + \left(1-\hat{\text{obj}}_i^l\right)\log\left(1-\text{obj}_i^l\right)\right) \tag{26}$$

where G means the number of grid cells, $\hat{\text{obj}}_i^l$ denotes the label (0 or 1) that whether the bounding box l of the sample i contains an object, and obj_i^l indicates the prediction probability for whether there is an object within this bounding box. Following FCOS [21] and YOLOX [22], the BCELoss function is used to train the classification branch and objectness branch, and the intersection over union (IoU) loss function is used to train for regression branch. λ is the balance factor of regression loss. We keep $\lambda = 5.0$ constant in all experiments.

IV. Experiments

In this section, we first describe the publicly available and challenging datasets. Then, the evaluation metrics and implementation details are provided for all experiments. Later, we prove the effectiveness of each proposed module in our ablation study. Finally, the detection results, feature extraction results, and comparisons with state-of-the-art methods are presented.

A. Datasets

1) RSOD [54]: It is a remote sensing dataset proposed by Wuhan University in 2017 covering aircraft, oil tank, overpass, and playground categories. It contains 2326 images and 6950 object instances (4493 aircraft, 1586 oil tanks, 191 playgrounds, and 180 overpasses). The image size of this dataset ranges from 512 × 512 to 1961 × 1193 pixels and its spatial resolutions vary from 0.3 to 3 m. We adopt the unified strategy in [55] to divide 25% images for training, 25% images for validation, and 50% for testing.

2) NWPU VHR-10: This dataset includes ten categories. It is proposed by Cheng et al. [56] of Northwestern Polytechnic University, Xi'an, China. Its latest version contains 1172 RSIs (400 × 400) pixels. Remote sensing objects are classified as airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor port, bridge, and vehicle. According to [9], we split 75% of the dataset as the training set and 25% of it as the testing set.

3) DIOR [8]: The DIOR dataset is a large-scale, publicly available benchmark for object detection in optical RSIs. It consists of 23 463 images (800 × 800 pixels) and 192 472 instances, covering 20 target classes: airplane (AL), airport (AT), baseball field (BF), basketball court (BC), bridge (BG), chimney (CM), dam (DM), expressway-service-area (EA), expressway-toll-station (ES), vehicle (VH), harbor (HB), overpass (OP), golf course (GC), ground track field (GF), ship (SP), stadium (SD), storage tank (ST), tennis court (TC), train station (TS), and windmill (WM). It has a spatial resolution of 0.5–30 m. There are 5862 and 5863 images in train and validation sets, respectively; then, the remaining 11 738 images are used as the test set.

B. Evaluation Metrics

We adopt the widely used average precision (AP) of each class and mean AP (mAP) of all classes as the evaluation metrics. A higher mAP generally indicates better detection performance. The AP and mAP are defined as follows:

$$\text{AP} = \int_0^1 P(R)dR \tag{27}$$

$$\text{mAP} = \frac{1}{N_{\text{cls}}}\sum_{i=1}^{N_{\text{cls}}}\text{AP}_i \tag{28}$$

where P and R represent the precision and the recall, respectively, and N_{cls} represents the total number of classes. The

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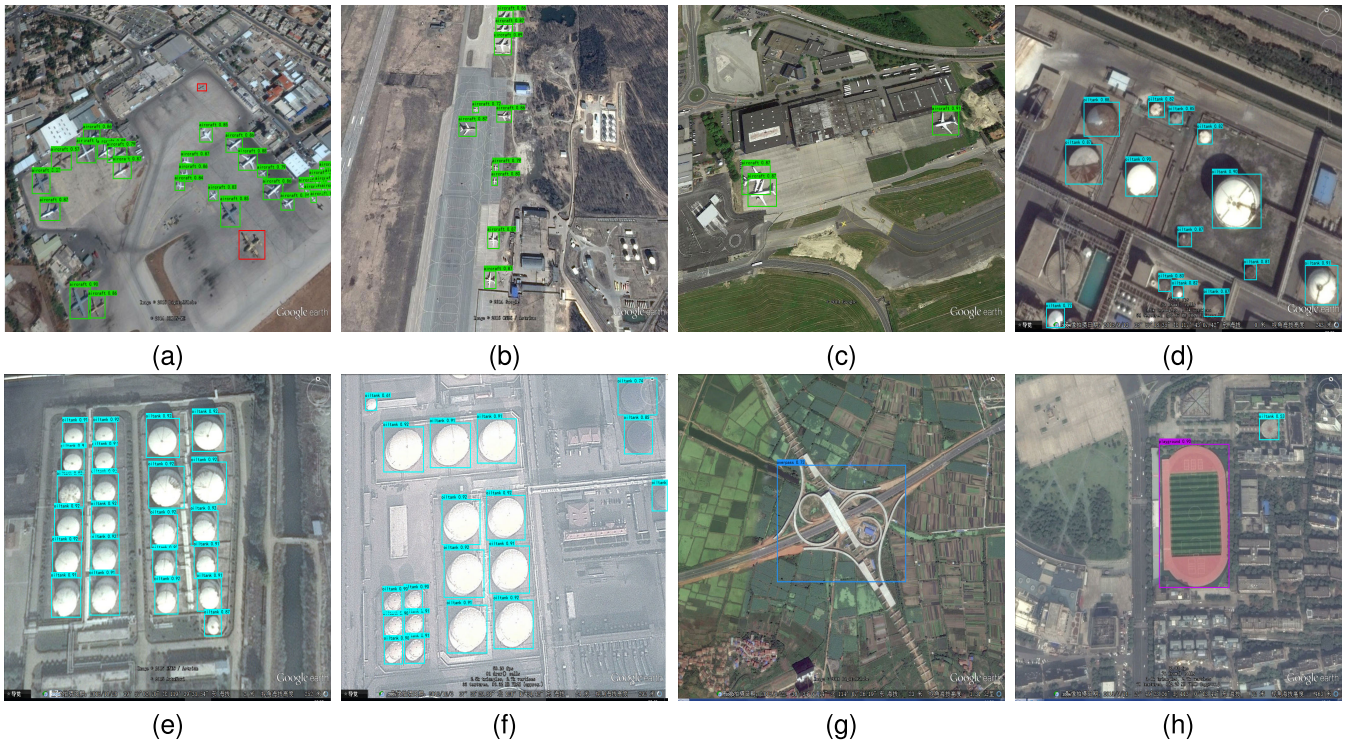


Fig. 8. Visualizing the detection results of our GLSANet on the RSOD dataset. The size of these representative images is uniform to facilitate an esthetic layout. (a)–(c) Aircraft. (d)–(f) Oil tank. (g) Overpass. (h) Playground. Red boxes are the missing predictions. Since the images we selected contain many small targets, zooming in on the page gives a better visual effect.

precision P and the recall R are expressed as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{29}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{30}$$

where TP, FP, and FN denote the number of true positives, false positives, and false negatives, respectively. A true positive satisfies two conditions: 1) the IoU between the predicted bounding box and ground truth is higher than 0.5 and 2) the predicted label is correct.

C. Implementation Details

Our experimental environment is the PyTorch framework (PyTorch1.6) in the Windows 10 operating system. All experiments are performed on two NVIDIA RTX4000 GPUs with 8-GB memory per GPU. We adopt the stochastic gradient descent (SGD) algorithm to optimize the parameters of the model. The initial learning rate of SGD is 0.01. The learning rate reduction strategy is a cosine annealing algorithm. The size of input images for the network is resized to 640×640 on three datasets. The batch size per GPU is 4. In addition, the mAP of all methods under the 0.5 IoU threshold is used to measure the detection results.

D. Ablation Study and Analysis

We perform an ablation experiment on RSOD to prove the effectiveness of GSIIM and LAPM. Table I shows the result of ablation experiment.

TABLE I
ABLATION EXPERIMENT ON RSOD [54]

Method	mAP	Aircraft	Oil tank	Overpass	Playground
Baseline	92.37	95.14	97.76	81.52	95.04
+GSIIM	94.43	96.14	97.93	88.43	95.21
+LAPM	94.69	96.76	98.13	85.74	98.13
+GSIIM+LAPM	95.93	96.80	98.46	91.45	97.01

1) Baseline: Our baseline is ResNet50 backbone with an FPN. We use ResNet50 to extract multiscale feature maps of RSIs. The FPN is applied to fuse features. In all experiments, all hyperparameters are consistent for a fair comparison. The baseline reaches 92.37% mAP on RSOD.

2) Effect of GSIIM: The proposed GSIIM is designed to excavate and reinforce the high-level semantic information in the deep feature map so that our network can recognize background and object clearly. As shown in Table I, our model (only with GSIIM) can reach 94.43% mAP (2.06% ↑). Each category has seen an increase on its corresponding AP. In particular, the AP of the aircraft category is improved by 1.00% and that of overpass category is improved by 6.91%. This illustrates that GSIIM really improves the network's understanding of the global information of images so that it can recognize the objects more clearly. Furthermore, it can also improve the detection accuracy of large-scale objects (e.g., overpass) since it works on the deep feature map.

3) Effect of LAPM: Considering that the features of small objects are likely to be submerged by background and noise, we introduce LAPM to amplify the features of

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TABLE II
COMPARISON BETWEEN THE GSIIM AND THE DATA AUGMENTATION WITH RANDOM ROTATION

Method	mAP	Aircraft	Oil tank	Overpass	Playground
Baseline	92.37	95.14	97.76	81.52	95.04
+DARR*	93.34	95.10	97.01	85.49	95.77
+GSIIM	94.43	96.14	97.93	88.43	95.21

DARR* means the data augmentation with random rotation.

them while suppressing the background and noise. The result shows that our method (only with LAPM) can reach 94.69% mAP (2.32% ↑) on RSOD. What is more, the AP of aircraft category, which contains many small objects, is also improved by 1.62%. It shows that LAPM really stands out the feature representation of small objects in the feature maps. In addition to aircraft, the APs of overpass and playground are also improved by 4.22% and 3.09% compared with the baseline, respectively. Moreover, when our method merges both GSIIM and LAPM together, we can see that the mAP, the APs of aircraft, oil tank, overpass, and playground are 3.56%, 1.66%, 0.7%, 9.93%, and 1.97% higher than those evaluated on baseline, respectively. This phenomenon demonstrates that the proposed GSIIM and LAPM are effective for multiscale RSOD. The utilization of them allows our approach to achieve outstanding detection results.

E. Comparison Between the GSIIM and the Data Augmentation With Random Rotation

To substantiate that the proposed GSIIM improves the object detection performance by extracting better global-aware information instead of augmenting the data into different rotated views, we conduct comparison experiments between the proposed GSIIM and the simple data augmentation with random rotations (DARR) on RSOD dataset [54]. We rotate each image in the training set by three random angles. These rotated images are expanded into this set. Next, we apply the augmented training set to train our baseline network. An original image and its three rotated versions are compared with one feature map and its three rotated versions. The experimental results are shown in Table II. DARR indeed enhances the object detection performance (0.97% ↑ in mAP) based on the baseline. However, we notice that DARR does not improve the detection performance on small-scale objects in our experiment, such as aircraft and oil tank (0.75% ↓). On the contrary, it performs well on large-scale objects, such as overpass (3.97% ↑) and playground (0.73% ↑). Although GSIIM obtains a lower AP in playground category than DARR (0.56% ↓), it achieves higher mAP and APs in the other three categories (aircraft, oil tank, and overpass), with improvements of 1.09% ↑, 1.04% ↑, 0.92% ↑, and 2.94% ↑ compared with DARR. Therefore, our proposed GSIIM achieves superior RSOD performance compared with DARR. The experimental results substantiate that GSIIM improves the object detection performance by extracting better global-aware information, instead of augmenting the data into different rotated views.

TABLE III
COMPARISON WITH STATE-OF-THE-ARTS ON RSOD [54]. THE BEST RESULTS ARE MARKED IN BOLD

Method	Backbone	mAP	Aircraft	Oil tank	Overpass	Playground
R-P-Faster RCNN	VGG16	84.47	70.84	90.19	78.74	98.09
Deformable R-FCN [57]	ResNet101	87.92	71.87	90.35	89.59	99.88
soft-NMS	VGG16	86.60	76.10	90.30	81.30	98.80
Sig-NMS [58]	VGG16	89.40	80.60	90.60	87.40	99.10
SSAFNet [59]	Hourglass104	92.82	95.75	98.39	84.66	92.50
ATTS	Hourglass104	91.30	94.50	96.00	74.90	**99.90**
Cascade R-CNN	ResNet101	91.30	94.20	96.10	83.20	99.00
DA^{2}FNet [55]	Hourglass104	94.78	95.72	97.73	90.56	95.12
GLSANet(ours)	ResNet50	**95.93**	**96.80**	**98.46**	**91.45**	97.01

F. Comparison Experiments and Analysis

The comparison experiments of GLSANet and other state-of-the-art detectors on RSOD [54], DIOR [8], and NWPU VHR-10 [56] are analyzed in this section. The results are shown in Tables III–V.

1) Results on RSOD: We compare our method with the advanced R-P-Faster RCNN and Deformable R-FCN in [57], soft-NMS and Sig-NMS in [58], SSAFNet in [59], and ATTS, Cascade R-CNN, and DA^{2}FNet in [55]. The result is shown in Table III. GLSANet reaches 95.93% mAP, which is the best performance among all competitors. This demonstrates the effectiveness of our proposed method. What is more, it also performed best on three classes: aircraft (96.80%), oil tank (98.46%), and overpass (91.45%). The AP of the small-scale aircraft is 1.08% higher than that of the second-place SSAFNet, which demonstrates that our method indeed improves the detection of small objects.

2) Results on DIOR: We evaluate the GLSANet on DIOR and compare it against the latest approaches, including CF2PN [60], CSFFNet [61], SRAF-Net [62], ASSD [63], RSADet [64], SB-MSN [65], and CoF-Net [66]. GLSANet is the only method that exceeds 77% mAP, as shown in Table IV. It achieves the best results in 14 of the total 20 categories: 95.8% AP on airplane, 92.9% AP on baseball field, 50.7% AP on bridge, 81.1% AP on chimney, 74.1% AP on expressway toll station, 87.6% AP on ground track field, 66.4% AP on harbor, 65.5% AP on overpass, 95.2% AP on ship, 92.4% AP on stadium, 86.3% AP on storage tank, 94.8% AP on tennis court, 62.1% AP on vehicle, and 89.2% AP on wind mill. This indicates that our proposed method does a good job to detect object at many different scales. Especially for categories that contain numerous small objects, such as airplane, ship, and vehicle, our GLSANet still performs best among all detectors. The AP on airplane (95.8%) is 7.4% higher than the second-place SRAF-Net. The AP on ship (95.2%) is 5.2% higher than the second-place RSADet. The AP on vehicle (62.1%) is 6.8% higher than the second-place RASDet.

3) Resluts on NWPU VHR-10: Table V compares the performance of our GLASNet with other state-of-the-art detectors on the NWPU VHR-10 dataset. The competitive methods include HRBM in [67], YOLOv3, Faster RCNN, FPN in [9], EGAT in [32], YOLOv7 in [68], and YOLOv8 in [69]. YOLOv7 and YOLOv8 are our implementation.

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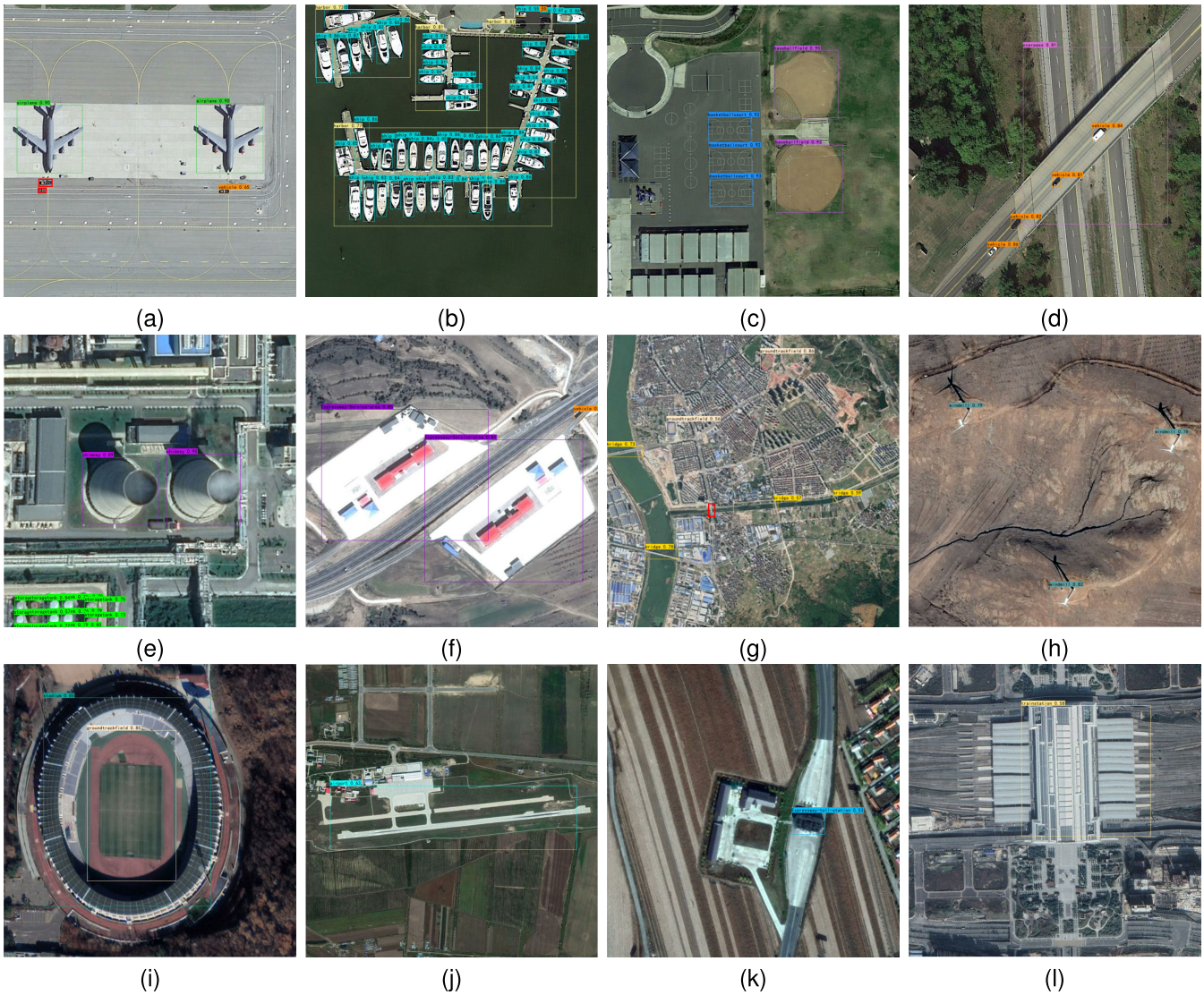


Fig. 9. Some representative detection results of GLSANet on the DIOR dataset. (a) Airplane and vehicle. (b) Ship, harbor, and vehicle. (c) Baseball field and basketball court. (d) Overpass and vehicle. (e) Chimney and storage tank. (f) Expressway service area and vehicle. (g) Ground track field and bridge. (h) Windmill. (i) Stadium and ground track field. (j) Airport. (k) Expressway toll station. (l) Train station. Red boxes are the missing predictions. Since the images we selected contain many small targets, zooming in on the page gives a better visual effect.

TABLE IV

COMPARISON WITH STATE-OF-THE-ARTS ON DIOR [8]. THE BEST RESULTS ARE MARKED IN BOLD

Method	Backbone	mAP	AL	AT	BF	BC	BG	CM	DM	EA	ES	GC	GF	HB	OP	SP	SD	ST	TC	TS	VH	WM
CF2PN [60]	VGG16	67.3	78.3	78.3	76.5	88.4	37.0	71.0	59.9	71.2	51.2	75.6	77.1	56.8	58.7	76.1	70.6	55.5	88.8	50.8	36.9	86.4
CSFFNet [61]	ResNet101	68.0	57.2	79.6	70.1	87.4	46.1	76.6	62.7	82.6	73.2	78.2	81.6	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9
SRAF-Net [62]	ResNet101	69.7	88.4	76.5	92.6	87.9	35.8	83.8	58.6	86.8	66.8	76.4	82.8	16.2	58.0	59.4	80.9	55.6	90.6	52.0	53.2	91.0
ASSD [63]	VGG16	71.1	85.6	82.4	75.8	89.5	40.7	77.6	64.7	67.1	61.7	80.8	78.6	62.0	58.0	84.9	76.7	65.3	87.9	62.4	44.5	76.3
RSADet [64]	DLA34+SAM	72.2	73.6	**86.0**	72.6	89.6	43.6	75.3	62.3	79.5	68.7	78.6	79.1	57.9	59.2	90.0	55.8	77.0	87.8	65.3	55.3	86.5
SB-MSN [65]	ResNeXt101	70.3	79.6	82.2	76.4	89.8	45.6	78.2	64.8	58.9	59.3	79.2	82.4	51.8	60.8	74.4	79.7	66.4	85.6	65.4	45.1	79.9
CoF-Net [66]	ResNet50	75.8	84.0	85.3	82.6	**90.0**	47.1	80.7	**73.3**	**89.3**	74.0	**84.5**	83.2	57.4	62.2	82.9	77.6	68.2	89.9	**68.7**	49.3	85.2
GLSANet(ours)	ResNet50	**77.9**	**95.8**	78.9	**92.9**	87.9	**50.7**	**81.1**	55.5	79.8	**74.1**	71.6	**87.6**	**66.4**	**65.5**	**95.2**	**92.4**	**86.3**	**94.8**	50.6	**62.1**	**89.2**

GLASNet reaches the best APs on four of ten categories, including the small-scale baseball diamond (99.40%) and tennis court (98.75%). It is the second place on another smaller scale ship class (95.80%). This demonstrates the robustness of our method on small object detection. However, we find that it is poor at detecting vehicles and only achieves 86.76% AP on this category. This is probably because these images containing vehicles are locally sharpened color infrared images rather than the RSIs normally taken by satellites. Thus, our GLSANet cannot adapt to these infrared images, which leads to its inferior performance on this class.

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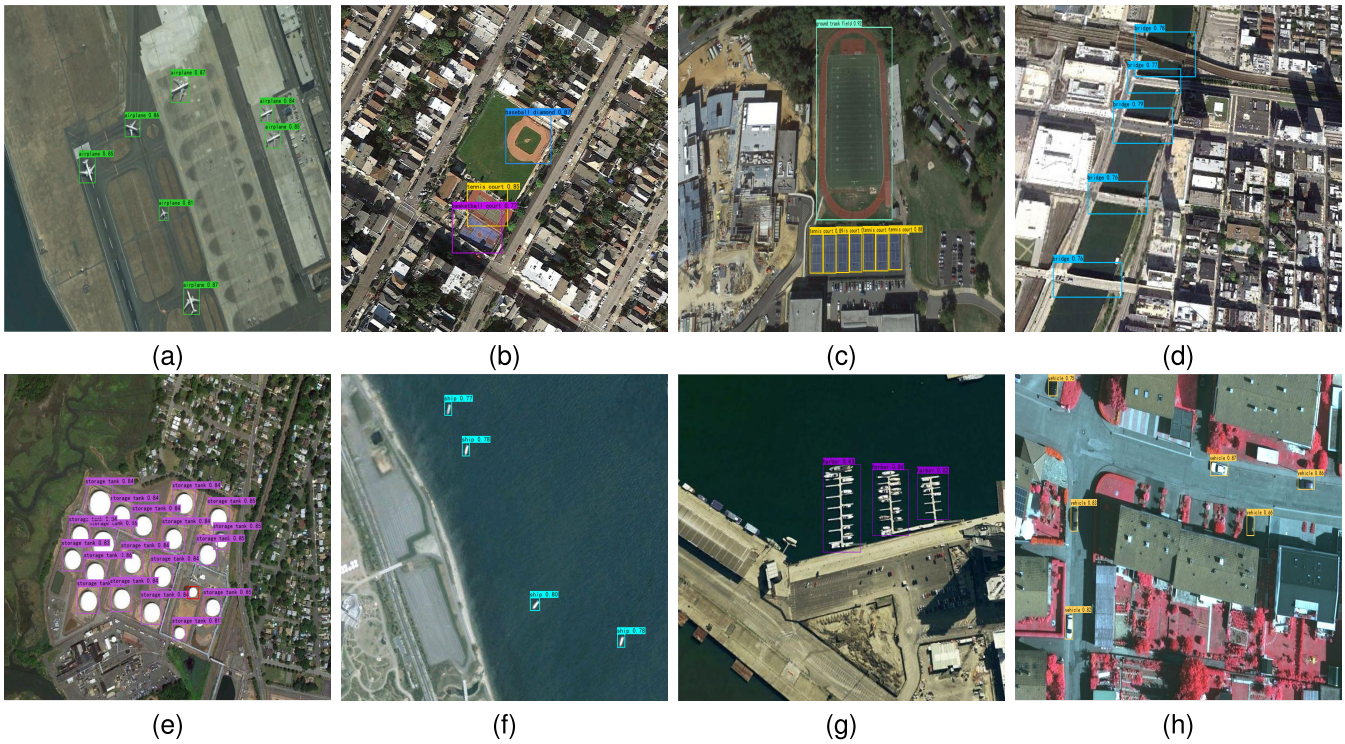


Fig. 10. Visualizing the detection results of our GLSANet on the NWPU VHR-10 dataset. (a) Airplane. (b) Baseball diamond, tennis court, and basketball court. (c) Ground track field and tennis court. (d) Bridge. (e) Storage tank. (f) Ship. (g) Harbor. (h) Vehicle. Red box in (e) is the missing prediction.

TABLE V
COMPARISON WITH STATE-OF-THE-ARTS ON NWPU VHR-10 [56]. THE BEST RESULTS ARE MARKED IN BOLD

Method	Backbone	mAP	Airplane	Ship	Storage tank	Baseball diamond	Tennis court	Basketball court	Ground trackfield	Harbor	Bridge	Vehicle
HRBM [67]	ZFCNN	87.12	99.70	90.80	90.61	92.91	90.29	80.13	90.81	80.29	68.53	87.14
YOLOv3	DarkNet53	87.27	99.55	81.28	80.30	98.26	80.56	81.82	99.47	74.31	89.61	86.98
Faster RCNN	ResNet50	88.30	**100**	85.28	**100**	95.93	87.59	92.08	99.73	92.11	43.37	80.60
FPN	ResNet50	90.80	**100**	90.86	99.99	96.84	90.67	**95.05**	**100**	93.67	50.86	90.19
EGAT [32]	GCN+LSTM	92.01	97.32	**96.72**	97.16	96.51	86.64	94.46	94.18	86.18	80.14	**90.79**
YOLOv7	DarkNet53	92.94	99.52	92.60	96.61	97.78	93.85	90.78	95.60	89.54	83.72	89.38
YOLOv8	DarkNet53	93.61	99.89	94.01	96.67	97.93	93.90	90.82	96.79	90.65	84.69	90.60
GLSANet(ours)	ResNet50	**94.57**	99.90	95.80	97.12	**99.40**	**98.75**	86.12	99.46	**97.51**	**84.83**	86.76

G. Detection Visualization Results

To further demonstrate the effectiveness of our proposed GLSANet, we visualize its prediction results on RSOD, DIOR, and NWPU VHR-10. As shown in Fig. 8, it achieves excellent detection performance on RSOD. Although there are two missing predictions in Fig. 8(a), almost all the small airplanes in Fig. 8(a)–(c) and the small oil tanks in Fig. 8(d) can be detected accurately. Figs. 9 and 10 show the prediction results of GLSANet on DIOR and NWPU VHR-10, respectively. We can see that although the RSIs of them contain large and complex background, our method still effectively detects the objects in them, especially the small-scale vehicles in Fig. 9(d) and (f), airplanes in Fig. 10(a), and ships in Fig. 10(f). Even though our method still has missing detections [e.g., Figs. 9(a) and 10(e)], its overall remarkable capability is still obvious. The visualization results suggest the superior performance of GSIIM and LAPM in multiscale object detection on RSIs again.

H. Feature Extraction Visualization Results

To further demonstrate the effects of GSIIM and LAPM on feature extraction, we visualize the feature maps of small-scale airplanes and ships in Fig. 11. Although the unprocessed feature maps in Fig. 11(b) are able to extract the features of these objects, there are still a lot of background features and noise in them (e.g., the features of curves on the ground of the airport and the sand beach on the left side of the harbors). This brings significant interference in small object detection. Moreover, the features of these ships are confused together. Therefore, we are unable to locate the feature of one ship accurately. As shown in Fig. 11(c), the background and noise are noticeably suppressed after we apply GSIIM. What is more, the features of these airplanes and ships become richer. Fig. 11(d) shows the changes of these features when we utilize LAPM based on GSIIM. It is obvious that these features become more clear and accurate. Even though the distribution of these ships is dense, the feature confusion

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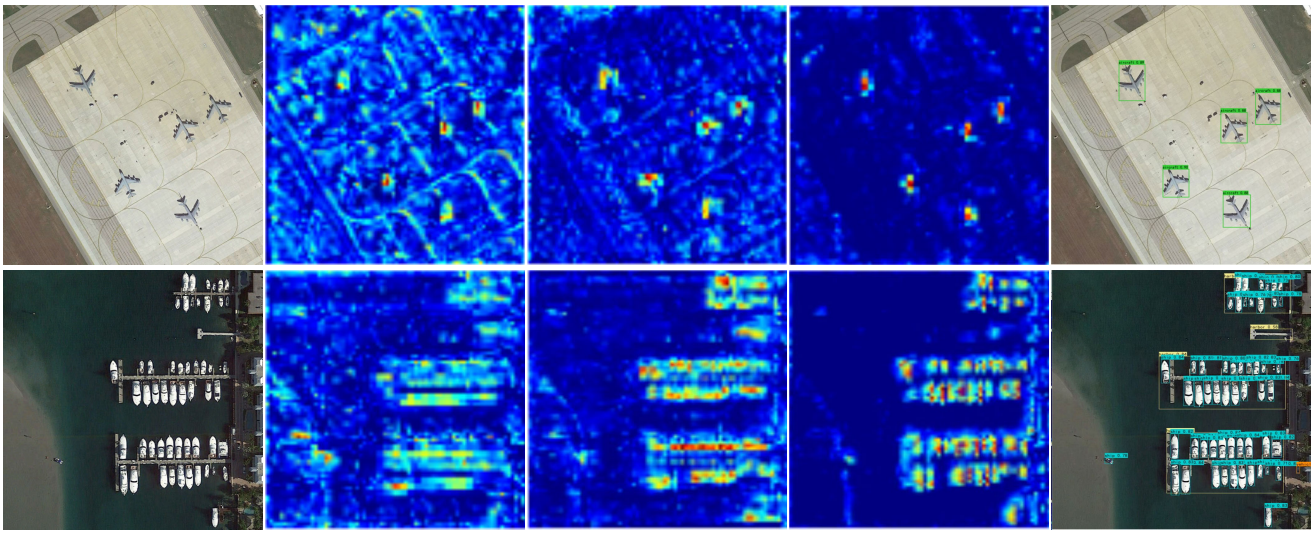


Fig. 11. Visualizing the feature extraction effects of GSIIM and LAPM. (a) Two original RSIs. (b) Feature maps without any processing. (c) Feature extraction results only with GSIIM. (d) Feature maps with both GSIIM and LAPM. (e) Final detection results.

between neighboring ships is greatly reduced with LAPM. The features of most ships are separated. The visualization results demonstrate that GSIIM and LAPM enhance the features of remote sensing objects (especially the small ones). They also reduce the influence of background and noise at the same time. These two modules provide better object detection performance.

V. Conclusion

In this article, a novel one-stage object detection network named GLSANet is proposed to improve the performance of multiscale object detection in RSIs, especially the small-scale objects. We design this network to effectively aggregate global semantic information and local feature information. First, the GSIIM is proposed to mine and enhance the high-level semantic information in the deep feature map. Second, the optimized FPN is applied to fuse features and realize multiscale object detection. Finally, the LAP is introduced to intensify the features of small objects contained in RSIs while diminishing background and noise. A series of comparison experiments with the same dataset settings on different detection methods proves the superior performance of the proposed GLASNet, especially in small object detection. Feature extraction visualization results demonstrate the working mechanisms of GSIIM and LAPM in our method. Our GLSANet may be able to provide some inspiration for the challenging problems of complex background and small object detection in RSIs. In addition, we find that the detection performance of our GLSANet on color infrared images is not satisfactory, which will be further investigated in the future.

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